

Step-by-Step During a GPR Survey: What to Expect on Site

A crew getting ready to cut concrete, core through a slab, trench across a site, drill anchors, or excavate near a building needs more than a plan on paper. Hidden utilities, conduits, rebar, post-tension cables, and voids can turn a simple task into an expensive problem. A GPR survey helps show what may be below concrete, asphalt, soil, or paved areas before work begins. For contractors, engineers, builders, and property owners, that information can reduce risk, limit delays, and help the job move forward with fewer surprises.

What Is A GPR Survey?

A [GPR survey](#) uses ground penetrating radar equipment to send radar signals into concrete, asphalt, soil, or other surfaces. When those signals are reflected back, a trained technician reads the data and looks for patterns that may indicate objects or changes beneath the surface.

Common findings can include rebar, post-tension cables, electrical conduits, water lines, sewer lines, communication lines, voids, unknown anomalies, and changes in concrete thickness. GPR concrete scanning is often used before drilling, cutting, coring, or anchoring because it can collect useful information from one side of a slab.

Step 1: What Happens Before The GPR Technician Arrives?

Before the site visit, the technician or project coordinator usually asks for basic details about the work. That may include the scan area, the type of work planned, known utility records, site drawings, access points, and safety requirements. A clear scope helps the technician focus GPR scanning efforts rather than wasting time on areas not part of the job. A clean and accessible work area also helps the visit run more smoothly.

Step 2: How Is The Survey Area Reviewed On Site?

The technician starts with a walk-through of the area. This step helps confirm access, surface conditions, and the exact locations that need to be scanned. Visible clues can matter, including utility covers, valves, floor penetrations, drains, conduit runs, patchwork, old markings, or nearby equipment. Site conditions are also checked at this stage. Rough terrain, heavy reinforcement, wet soil, tight rooms, parked equipment, and blocked access can affect how the work is handled.

Step 3: How Does GPR Scanning Actually Work?

During GPR scanning, the technician moves the radar equipment across the target area in controlled passes. The system collects reflected signals and displays the data on a screen. The equipment does not automatically name each pipe, cable, void, or piece of steel. A skilled technician reads the signal patterns and compares them with site clues. Think of it less like taking a photo and more like reading a set of traces that need trained eyes.

Step 4: What Does The Technician Look For During The GPR Survey?

The technician looks for patterns that may suggest utilities, conduits, rebar, post-tension cables, tanks, voids, or other subsurface changes. Outdoor utility locating may also call for electromagnetic locating equipment, especially when conductive utilities are present. Soil type, depth, moisture, target material, and below-grade congestion can all affect results. For larger jobs, those findings may later support GPR utility mapping, giving the project team a better record of what was found on site.

Step 5: How Are Findings Marked After The GPR Inspection?

Once likely subsurface targets are identified, the technician may mark the surface with paint, chalk, flags, tape, or another agreed method. These markings help crews see where caution is needed before cutting, coring, trenching, drilling, or excavating. Good GPR inspection services should leave the site team with markings that are clear enough to use during active work, especially when several trades are moving through the same area.

Step 6: What Is Included In GPR Utility Mapping?

For some projects, field markings are only part of the deliverable. The findings may also be converted into sketches, reports, photos, site maps, CAD files, KMZ files, or other documentation, depending on the project scope. GPR utility mapping can be useful for large properties, renovation planning, facility maintenance, engineering coordination, and future repairs. A clear map can also help reduce the need for repeated locating later.

Step 7: What Should You Do With The Survey Results?

A GPR survey gives the project team better information, but safe work practices still matter. Crews should review markings and maps before work starts, share the findings with the right trades, and compare the results with drawings, permits, and site safety procedures. GPR can reduce uncertainty, but no scanning method can guarantee that every object will be found under all conditions.

When Should You Schedule A GPR Survey?

A GPR survey is best scheduled before work begins, especially when hidden utilities, reinforcement, or other subsurface features could affect safety, cost, or timing. Scanning first is usually easier than dealing with a damaged utility, a stopped crew, or an emergency repair.

- Concrete cutting or coring
- Trenching or excavation
- Renovation work
- Utility installation
- Structural investigation
- Site planning
- Private utility locating
- Roadway, parking lot, or slab work

Common Questions About A GPR Survey

Most questions come down to timing, site conditions, and the level of confidence the scan can provide before work starts.

How Long Does A GPR Survey Take?

The timeline depends on the scan area, site access, surface conditions, and reporting needs. A small, open area may move quickly. A larger site with several scan zones, heavy reinforcement, or detailed mapping requirements can take longer.

Can GPR Find Every Underground Utility?

GPR is useful, but results depend on soil conditions, depth, material type, moisture, and site complexity. Non-metallic utilities, wet clay, deep targets, and crowded underground areas can be harder to interpret. Experience matters because the data needs careful review, not guesswork.

Is GPR Scanning Safe For Active Jobsites?

Yes, [GPR scanning](#) is non-destructive and does not require digging into the surface during the scan. The technician moves the equipment across the target area and reviews reflected radar data, making it practical for active jobsites that need information before opening a slab or ground surface.

Do I Need GPR Inspection Services Before Drilling Into Concrete?

GPR inspection services are strongly recommended when hidden conduits, rebar, post-tension cables, or other embedded objects may be present. Drilling without checking first can damage

utilities, create safety concerns, and delay the job. A scan helps crews understand where caution is needed before drilling, anchoring, coring, or cutting.

Plan Safer Work With A Professional GPR Survey

A GPR survey helps contractors, engineers, builders, and property owners see what may be hidden below the surface before work begins. From site review and GPR scanning to field markings and GPR utility mapping, the process supports safer planning and fewer avoidable setbacks. Superior Scanning is a practical service choice for projects that need clear subsurface information before cutting, coring, drilling, trenching, or excavating.